Φάση Ε: Model Evaluation & Σύγκριση

Υπεύθυνος: ML Engineer

Μετρικές:

- Accuracy
- Precision / Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

Οπτικοποιήσεις:

1. Confusion Matrix (heatmap)
2. ROC Curve
3. Συγκριτικά bar charts
4. Πίνακας σύγκρισης

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# Φάση Ε & ΣΤ: Model Evaluation, Σύγκριση & Συμπεράσματα (ΠΛΗΡΗΣ ΚΑΛΥΨΗ + F1-
SCORE)
from pyspark.sql import SparkSession
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, confusion_matrix, roc_curve, auc,
                             precision_recall_fscore_support,
classification_report)

print("Εκκίνηση SparkSession...")
spark =
SparkSession.builder.appName("Evaluation_Metrics_Final").master("local[*]").getOrCreate()

print("Φόρτωση προβλέψεων (Pandas)...")
df_rf = spark.read.parquet("../data/rf_predictions.parquet").toPandas()
df_svm = spark.read.parquet("../data/svm_predictions.parquet").toPandas()
df_nb = spark.read.parquet("../data/nb_predictions.parquet").toPandas()
df_mlp = spark.read.parquet("../data/mlp_predictions.parquet").toPandas()
df_gbt = spark.read.parquet("../data/gbt_predictions.parquet").toPandas()
df_lr = spark.read.parquet("../data/lr_predictions.parquet").toPandas()

# Λεξικό για πλήρη αποθήκευση μετρικών ανά κλάση και συνολικά
metrics_dict = {
    "Model": [],
    "Acc (Total)": [],
    "P (Class 0)": [], "R (Class 0)": [], "F1 (Class 0)": [],
    "P (Class 1)": [], "R (Class 1)": [], "F1 (Class 1)": [],
    "ROC-AUC": []
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}

def evaluate_model_full(df, model_name, prob_col="probability"):
    y_true = df['stroke'].values
    y_pred = df['prediction'].values

    # ROC-AUC
    y_prob = np.array([v[1] for v in df[prob_col]]) if prob_col == "probability"
else np.array([v[1] for v in df[prob_col]])
    fpr, tpr, _ = roc_curve(y_true, y_prob)
    roc_auc = auc(fpr, tpr)

    # Μετρικές ανά κλάση (Precision, Recall, F1)
    precisions, recalls, f1s, _ = precision_recall_fscore_support(y_true, y_pred,
labels=[0, 1], zero_division=0)
    acc = accuracy_score(y_true, y_pred)

    # Προσθήκη στο λεξικό
    metrics_dict["Model"].append(model_name)
    metrics_dict["Acc (Total)"].append(acc)
    metrics_dict["P (Class 0)"].append(precisions[0])
    metrics_dict["R (Class 0)"].append(recalls[0])
    metrics_dict["F1 (Class 0)"].append(f1s[0])
    metrics_dict["P (Class 1)"].append(precisions[1])
    metrics_dict["R (Class 1)"].append(recalls[1])
    metrics_dict["F1 (Class 1)"].append(f1s[1])
    metrics_dict["ROC-AUC"].append(roc_auc)

    cm = confusion_matrix(y_true, y_pred)

    # Εκτύπωση αναλυτικού classification report
    print(f"\n{'='*20} {model_name} {'='*20}")
    print(classification_report(y_true, y_pred, target_names=["Class 0", "Class
1"]))

    return fpr, tpr, roc_auc, cm

# Εκτέλεση αξιολόγησης (Τα 6 Τελικά Μοντέλα μας)
fpr_rf, tpr_rf, auc_rf, cm_rf = evaluate_model_full(df_rf, "Random Forest",
"probability")
fpr_svm, tpr_svm, auc_svm, cm_svm = evaluate_model_full(df_svm, "SVM",
"rawPrediction")
fpr_nb, tpr_nb, auc_nb, cm_nb = evaluate_model_full(df_nb, "Naive Bayes",
"probability")
fpr_mlp, tpr_mlp, auc_mlp, cm_mlp = evaluate_model_full(df_mlp, "Neural
Networks", "probability")
fpr_gbt, tpr_gbt, auc_gbt, cm_gbt = evaluate_model_full(df_gbt, "GBT Classifier",
"probability")
fpr_lr, tpr_lr, auc_lr, cm_lr = evaluate_model_full(df_lr, "Logistic Reg.",
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"probability")


# ============================================================
# 1. ΠΙΝΑΚΑΣ ΣΥΓΚΡΙΣΗΣ (Κεφάλαιο 4: Model Evaluation)
# ============================================================
print("\n" + "="*80)
print("ΠΙΝΑΚΑΣ ΣΥΓΚΡΙΣΗΣ ΜΟΝΤΕΛΩΝ (Απαίτηση Εκφώνησης)")
print("="*80)
comparison_df = pd.DataFrame(metrics_dict).round(3)
print(comparison_df.to_string(index=False))
print("="*80)


# ============================================================
# 2. ΟΠΤΙΚΟΠΟΙΗΣΗ 1: CONFUSION MATRICES (Τέλειο 2x3 Grid)
# ============================================================
fig, axes = plt.subplots(2, 3, figsize=(16, 10))
fig.suptitle('Confusion Matrices (Απαίτηση Εκφώνησης)', fontsize=18,
fontweight='bold')

cms = [cm_rf, cm_svm, cm_nb, cm_mlp, cm_gbt, cm_lr]
titles = ['Random Forest', 'SVM', 'Naive Bayes', 'Neural Networks', 'GBT
Classifier', 'Logistic Reg.']
colors = ['Blues', 'Oranges', 'Greens', 'Purples', 'Reds', 'YlGnBu']

for i in range(len(colors)):
    row, col = i // 3, i % 3
    sns.heatmap(cms[i], annot=True, fmt='d', cmap=colors[i], cbar=False,
ax=axes[row, col])
    axes[row, col].set_title(titles[i], fontsize=12)
    axes[row, col].set_xlabel('Predicted')
    axes[row, col].set_ylabel('True')

plt.tight_layout()
plt.subplots_adjust(top=0.9)
plt.show()

# ============================================================
# 3. ΟΠΤΙΚΟΠΟΙΗΣΗ 2: ROC CURVES
# ============================================================
plt.figure(figsize=(11, 8))
plt.plot(fpr_rf, tpr_rf, label=f'Random Forest (AUC = {auc_rf:.3f})', lw=2)
plt.plot(fpr_svm, tpr_svm, label=f'SVM (AUC = {auc_svm:.3f})', lw=2)
plt.plot(fpr_nb, tpr_nb, label=f'Naive Bayes (AUC = {auc_nb:.3f})', lw=2)
plt.plot(fpr_mlp, tpr_mlp, label=f'Neural Networks (AUC = {auc_mlp:.3f})', lw=2)
plt.plot(fpr_gbt, tpr_gbt, label=f'GBT (AUC = {auc_gbt:.3f})', lw=2,
linestyle='--')
plt.plot(fpr_lr, tpr_lr, label=f'Logistic Reg. (AUC = {auc_lr:.3f})', lw=2,
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linestyle='--')
plt.plot([0, 1], [0, 1], 'k--', lw=2, label='Random Chance')
plt.title('Σύγκριση Μοντέλων: Καμπύλη ROC', fontsize=16, fontweight='bold')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate (Recall)')
plt.legend(loc="lower right")
plt.grid(alpha=0.3)
plt.show()

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# 4. ΟΠΤΙΚΟΠΟΙΗΣΗ 3: ΣΥΓΚΡΙΤΙΚΑ BAR CHARTS
# ==========================================================
bar_df = comparison_df.drop(columns=["ROC-AUC"])
metrics_melted = bar_df.melt(id_vars="Model", var_name="Metric",
value_name="Score")

plt.figure(figsize=(14, 7))
sns.barplot(x='Model', y='Score', hue='Metric', data=metrics_melted,
palette='Dark2')
plt.title('Συγκριτικό Διάγραμμα Μετρικών (Accuracy, Precision, Recall, F1-
Score)', fontsize=16, fontweight='bold')
plt.ylim(0, 1.1)
plt.ylabel('Score')
plt.xticks(rotation=30)
plt.legend(title='Μετρικές', bbox_to_anchor=(1.01, 1), loc='upper left')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()

spark.stop()
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Εκκίνηση SparkSession...
Φόρτωση προβλέψεων (Pandas)...

==================== Random Forest ====================
              precision    recall  f1-score   support

     Class 0       0.98      0.78      0.87       945
     Class 1       0.11      0.60      0.18        42

    accuracy                           0.77       987
   macro avg       0.54      0.69      0.53       987
weighted avg       0.94      0.77      0.84       987


==================== SVM ====================
              precision    recall  f1-score   support

     Class 0       0.98      0.75      0.85       945
     Class 1       0.10      0.64      0.17        42
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    accuracy                           0.74       987
   macro avg       0.54      0.69      0.51       987
weighted avg       0.94      0.74      0.82       987


==================== Naive Bayes ====================
              precision    recall  f1-score   support

     Class 0       0.99      0.37      0.54       945
     Class 1       0.06      0.88      0.11        42

    accuracy                           0.39       987
   macro avg       0.52      0.63      0.32       987
weighted avg       0.95      0.39      0.52       987


==================== Neural Networks ====================
              precision    recall  f1-score   support

     Class 0       0.98      0.76      0.85       945
     Class 1       0.10      0.62      0.18        42

    accuracy                           0.75       987
   macro avg       0.54      0.69      0.52       987
weighted avg       0.94      0.75      0.83       987


==================== GBT Classifier ====================
              precision    recall  f1-score   support

     Class 0       0.98      0.77      0.86       945
     Class 1       0.11      0.67      0.19        42

    accuracy                           0.76       987
   macro avg       0.55      0.72      0.53       987
weighted avg       0.94      0.76      0.83       987


==================== Logistic Reg. ====================
              precision    recall  f1-score   support

     Class 0       0.98      0.75      0.85       945
     Class 1       0.10      0.64      0.18        42

    accuracy                           0.74       987
   macro avg       0.54      0.70      0.51       987
weighted avg       0.94      0.74      0.82       987
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ΠΙΝΑΚΑΣ ΣΥΓΚΡΙΣΗΣ ΜΟΝΤΕΛΩΝ (Απαίτηση Εκφώνησης)

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Model	Acc (Total)	P (Class 0)	R (Class 0)	F1 (Class 0)	P (Class 1)	R (Class 1)	F1 (Class 1)	ROC-AUC
Random Forest	0.774	0.978	0.782	0.869	0.108	0.595	0.183	0.776
SVM	0.742	0.979	0.746	0.847	0.101	0.643	0.175	0.766
Naive Bayes	0.392	0.986	0.370	0.538	0.059	0.881	0.110	0.723
Neural Networks	0.753	0.978	0.759	0.855	0.102	0.619	0.176	0.750
GBT Classifier	0.764	0.981	0.768	0.862	0.113	0.667	0.194	0.780
Logistic Reg.	0.745	0.979	0.749	0.849	0.102	0.643	0.176	0.774

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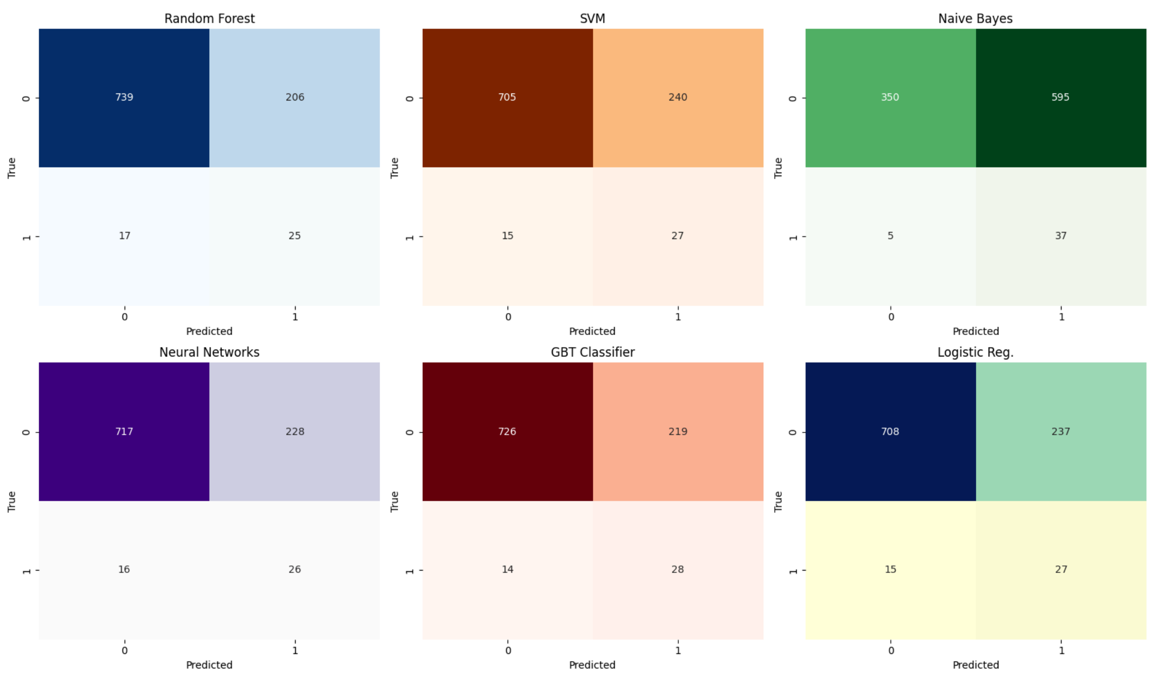
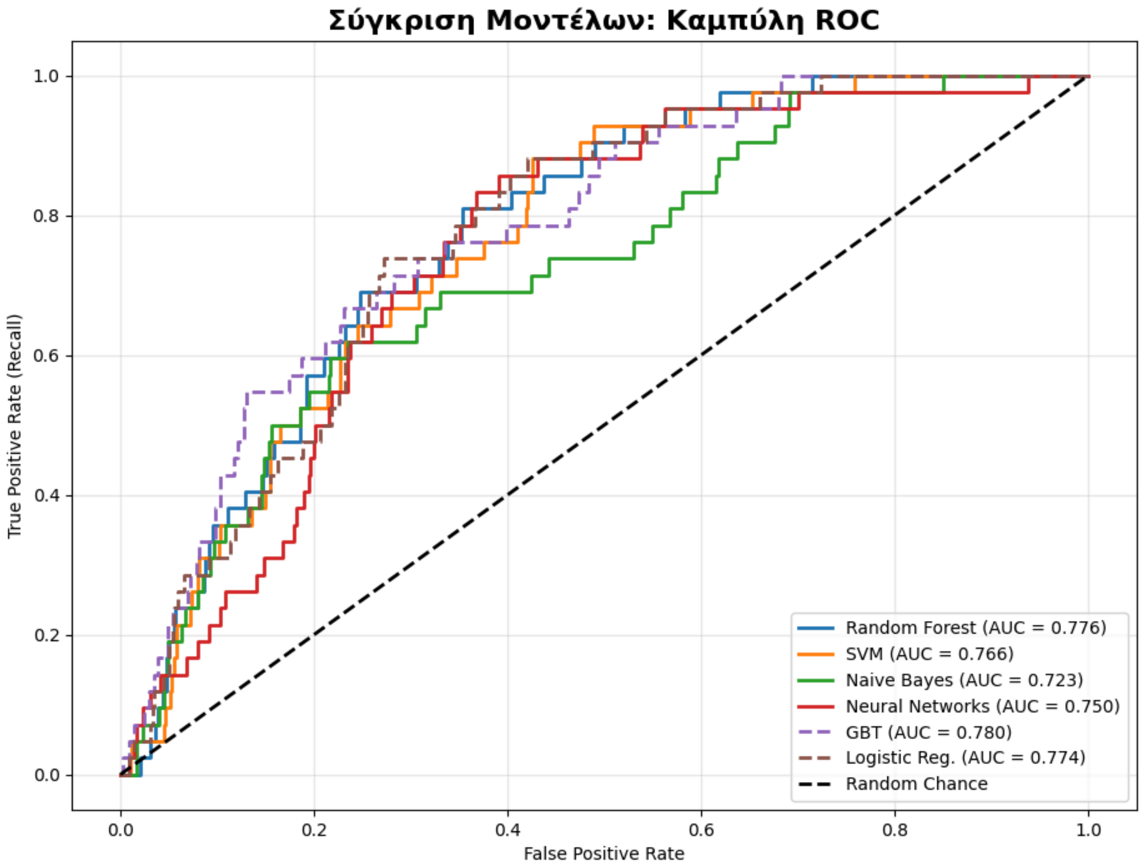
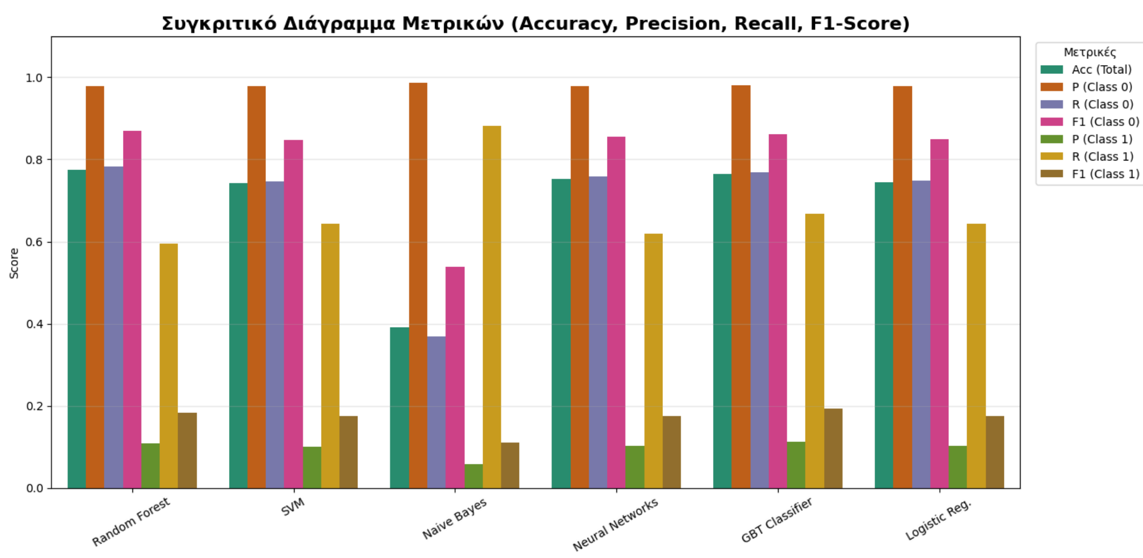
Confusion Matrices (Απαίτηση Εκφώνησης)




Σύγκριση Μοντέλων: Καμπύλη ROC
True Positive Rate (Recall)
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Random Forest (AUC = 0.776)
SVM (AUC = 0.766)
Naive Bayes (AUC = 0.723)
Neural Networks (AUC = 0.750)
GBT (AUC = 0.780)
Logistic Reg. (AUC = 0.774)
Random Chance


Συγκριτικό Διάγραμμα Μετρικών (Accuracy, Precision, Recall, F1-Score)
Score
Model
0.0
0.2
0.4
0.6
0.8
1.0
Random Forest
SVM
Naive Bayes
Neural Networks
GBT Classifier
Logistic Reg.
Μετρικές
Acc (Total)
P (Class 0)
R (Class 0)
F1 (Class 0)
P (Class 1)
R (Class 1)
F1 (Class 1)